

Brief Solutions

Problem 1. Conceptual Questions. Write an equation, an expression, or a short sentence to describe each of these concepts. [5 marks]

(a) The line integral $\int_C \vec{F} \cdot d\vec{r}$ is **path-independent** in a domain D .

Solution. For every pair A, B in D , the value of $\int_C \vec{F} \cdot d\vec{r}$ is the same for all paths from A to B .

Other acceptable answers: $\vec{F} = \vec{\nabla} f$ for scalar f ; $\oint_C \vec{F} \cdot d\vec{r} = 0$ for all closed loops in D ; $\vec{\nabla} \times \vec{F} = \vec{0}$ and D is simply connected. □

(b) The **exactness criteria** on $\vec{F}$ for the differential form $\vec{F} \cdot d\vec{r}$ to be an exact Pfaffian df for some scalar potential f .

Solution.

$$\vec{\nabla} \times \vec{F} = \vec{0}$$

□

(c) A vector field $\vec{F}$ is **incompressible**.

Solution.

$$\vec{\nabla} \cdot \vec{F} = 0$$

□

(d) A scalar field f is the **scalar potential** of a vector field $\vec{F}$.

Solution.

$$\vec{F} = \vec{\nabla} f$$

□

(e) The **directional derivative** of a scalar field f in the direction of a unit vector $\hat{u}$.

Solution.

$$D_{\hat{u}} f = \vec{\nabla} f \cdot \hat{u}$$

□

Problem 2. Find $\vec{\nabla} f$ and its value at P for

$$f = \frac{x}{x^2 + y^2}, \quad P(1, 1).$$

[5 marks]

Solution.

$$\frac{\partial}{\partial x} \left(\frac{x}{x^2 + y^2} \right) = \frac{(x^2 + y^2)(\partial_x x) - x(\partial_x (x^2 + y^2))}{(x^2 + y^2)^2} = \frac{(x^2 + y^2) - x(2x)}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

$$\frac{\partial}{\partial y} \left(\frac{x}{x^2 + y^2} \right) = x(-1)(x^2 + y^2)^{-2}(2y) = \frac{2xy}{(x^2 + y^2)^2}$$

□

Problem 3. Find the **divergence** of the following vector field $\vec{v}$ at the point $P(1, 0, -1)$:

$$\vec{v} = \frac{1}{(x^2 + y^2 + z^2)^{3/2}} (x\mathbf{i} + y\mathbf{j} + z\mathbf{k}).$$

[5 marks]

Solution.

$$\begin{aligned} \frac{\partial}{\partial x} \frac{x}{(x^2 + y^2 + z^2)^{3/2}} &= \frac{(x^2 + y^2 + z^2)^{3/2} (\partial_x x) - x ((x^2 + y^2 + z^2)^{3/2})'}{[(x^2 + y^2 + z^2)^{3/2}]^2} \\ &= \frac{(x^2 + y^2 + z^2)^{3/2} - x (3/2 (x^2 + y^2 + z^2)^{1/2} (2x))}{(x^2 + y^2 + z^2)^3} \\ &= \frac{(x^2 + y^2 + z^2)^{3/2} [x^2 + y^2 + z^2 - 3x^2]}{(x^2 + y^2 + z^2)^3} = \frac{y^2 + z^2 - 2x^2}{(x^2 + y^2 + z^2)^{5/2}} \end{aligned}$$

By Symmetry:

$$\vec{\nabla} \cdot \vec{v} = \frac{(y^2 + z^2 - 2x^2) + (x^2 + z^2 - 2y^2) + (y^2 + x^2 - 2z^2)}{(x^2 + y^2 + z^2)^{5/2}} = 0$$

□

Problem 4. Suppose $\vec{u} = u_1(x, y, z)\mathbf{i} + u_2(x, y, z)\mathbf{j} + u_3(x, y, z)\mathbf{k}$ and $\vec{v} = v_1(x, y, z)\mathbf{i} + v_2(x, y, z)\mathbf{j} + v_3(x, y, z)\mathbf{k}$. Prove the following formula:

$$\nabla \cdot (\vec{u} \times \vec{v}) = \vec{v} \cdot (\nabla \times \vec{u}) - \vec{u} \cdot (\nabla \times \vec{v}).$$

Hint: Use the triple scalar product formula:

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

[5 marks]

Solution.

$$\begin{aligned} \vec{\nabla} \cdot (\vec{u} \times \vec{v}) &= \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \frac{\partial}{\partial x} [u_2 v_3 - u_3 v_2] - \frac{\partial}{\partial y} [u_1 v_3 - u_3 v_1] + \frac{\partial}{\partial z} [u_1 v_2 - u_2 v_1] \\ &= [(u_2)_x v_3 + u_2 (v_3)_x] - [(u_3)_x v_2 + u_3 (v_2)_x] - [(u_1)_y v_3 + u_1 (v_3)_y] + [(u_3)_y v_1 + u_3 (v_1)_y] \\ &\quad + [(u_1)_z v_2 + u_1 (v_2)_z] - [(u_2)_z v_1 + u_2 (v_1)_z] \\ &= u_1 \left[\frac{\partial}{\partial z} v_2 - \frac{\partial}{\partial y} v_3 \right] + u_2 \left[\frac{\partial}{\partial x} v_3 - \frac{\partial}{\partial z} v_1 \right] + u_3 \left[\frac{\partial}{\partial y} v_1 - \frac{\partial}{\partial x} v_2 \right] \\ &\quad - v_1 \left[\frac{\partial}{\partial z} u_2 - \frac{\partial}{\partial y} u_3 \right] - v_2 \left[\frac{\partial}{\partial x} u_3 - \frac{\partial}{\partial z} u_1 \right] - v_3 \left[\frac{\partial}{\partial y} u_1 - \frac{\partial}{\partial x} u_2 \right] \\ &= -\vec{u} \cdot (\vec{\nabla} \times \vec{v}) + \vec{v} \cdot (\vec{\nabla} \times \vec{u}) = \vec{v} \cdot (\vec{\nabla} \times \vec{u}) - \vec{u} \cdot (\vec{\nabla} \times \vec{v}) \end{aligned}$$

□

Problem 5. Given that the differential form under the integral is exact (you don't need to show this), find the result of the following line-integral:

$$\int_{(0,0,0)}^{(1,1,0)} e^{x^2+y^2+z^2} (x dx + y dy + z dz).$$

[5 marks]

Solution. Set up: (i) $\frac{\partial f}{\partial x} = x e^{x^2+y^2+z^2}$; (ii) $\frac{\partial f}{\partial y} = y e^{x^2+y^2+z^2}$; (iii) $\frac{\partial f}{\partial z} = z e^{x^2+y^2+z^2}$

from (i): $f(x,y,z) = \int_{(y,z)} \frac{\partial f}{\partial x} dx = \frac{1}{2} e^{x^2+y^2+z^2} + g(y,z)$ using substitution $u = x^2+y^2+z^2$ and $du = 2x dx$

then $\frac{\partial f}{\partial y} = y e^{x^2+y^2+z^2} + \frac{\partial g}{\partial y}$. Compare with (ii) to get $\frac{\partial g}{\partial y} = 0 \Rightarrow g(y,z) = \int_{(z)} \frac{\partial g}{\partial y} dy = h(z)$

then $f(x,y,z) = \frac{1}{2} e^{x^2+y^2+z^2} + h(z)$

so $\frac{\partial f}{\partial z} = z e^{x^2+y^2+z^2} + h'(z)$. Compare with (iii) to get $h'(z) = 0 \Rightarrow h(z) = \text{Const}$ (set to zero)

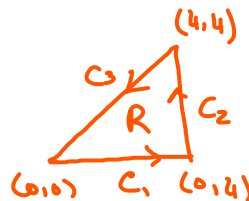
Thus $f(x,y,z) = \frac{1}{2} e^{x^2+y^2+z^2}$

$$\text{So } \int_{(0,0,0)}^{(1,1,0)} e^{x^2+y^2+z^2} (x dx + y dy + z dz) = f(1,1,0) - f(0,0,0) = \frac{1}{2} e^2 - \frac{1}{2} e^0 = \frac{1}{2} (e^2 - 1). \quad \square$$

Problem 6. Calculate the area of the triangular region with vertices $(0,0)$, $(0,4)$, $(4,4)$ using *only line integrals* along the boundary of the right-angled triangle.

[5 marks]

Solution. We know: $A = \frac{1}{2} \oint_C (x dy - y dx)$
 $= \frac{1}{2} \int_{C_1} x dy - y dx + \frac{1}{2} \int_{C_2} x dy - y dx + \frac{1}{2} \int_{C_3} x dy - y dx$



Parametrize:

$C_1: x=t, 0 \leq t \leq 4, y=0$ $C_2: x=4, 0 \leq t \leq 4, y=t$ $C_3: x=4-t, 0 \leq t \leq 4, y=4-t$

$$\text{Then } A = \frac{1}{2} \left[\int_0^4 \left(\underset{x \frac{dy}{dt}}{t \cdot 0} - \underset{-y \frac{dx}{dt}}{0 \cdot 1} \right) dt + \int_0^4 (4 \cdot (1) - t \cdot (0)) dt + \int_0^4 [(4-t)(-1) - (4-t)(-1)] dt \right]$$

$$A = \frac{1}{2} \left[0 + 4t \Big|_0^4 + 0 \right] = \frac{1}{2} (16) = 8. \quad \square$$